MVC

MVC Composite Supplemental Material-Value Tables

Density · CTE · Dk · Df · Thermal Conductivity

Fracture Toughness · PSD · Laser Absorption · Benchmarks

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1. Density & Physical Properties

Property	Symbol	Value	Unit	Std. Uncertainty (k=2)	Test Method
Bulk density	ρ	1.68	g/cm³	±0.03	ASTM D792
Apparent density (green)	ρ_a	1.72	g/cm³	±0.04	ASTM D792
True density (pycnometry)	ρ_t	2.14	g/cm³	±0.02	ASTM D5965
Porosity (total)	ϕ	21.5	%	±1.5	Mercury intrusion
Open porosity	ϕ_o	3.2	%	±0.5	Mercury intrusion
Closed porosity	ϕ_c	18.3	%	±1.2	Calc. from ρ
Water absorption (24h)	W_a	0.08	%	±0.01	IPC-TM-650 2.6.2
Moisture Dk coefficient	MCD	0.002	per% RH	±0.0003	IPC-TM-650 2.5.5.5

Table 1-1: Density and physical properties; k=2 expanded uncertainty (95% confidence).

2. Coefficient of Thermal Expansion (CTE)

Direction	Temp. Range	CTE	Unit	Uncert.	Method
X (in-plane)	-55 to +150 C	14.2	ppm/C	±1.0	TMA / IPC 2.4.41
Y (in-plane)	-55 to +150 C	14.5	ppm/C	±1.0	TMA / IPC 2.4.41
Z (through, below Tg)	-55 to Tg	42	ppm/C	±4	TMA / IPC 2.4.41
Z (through, above Tg)	Tg to +260 C	168	ppm/C	±10	TMA / IPC 2.4.41
Biaxial avg X-Y	25 to +125 C	14.3	ppm/C	±0.8	TMA, 10 C/min
Dk temp coeff. (TDk)	-40 to +85 C	-8	ppm/C	±1	Cavity resonance, variable T

Table 2-1: CTE data from TMA at 10 C/min heating rate.

3. Dielectric Constant (Dk) & Dissipation Factor (Df)

Freq.	Dk	Dk Uncert.	Df (tan d)	Df Uncert.	Method	Condition
1 GHz	3.10	±0.05	0.0028	±0.0003	IPC 2.5.5.5C	23C/50%RH
2 GHz	3.08	±0.05	0.0029	±0.0003	IPC 2.5.5.5C	23C/50%RH
5 GHz	3.05	±0.05	0.0031	±0.0003	IPC 2.5.5.5C	23C/50%RH
10 GHz	2.98	±0.05	0.0034	±0.0004	Cavity res.	23C/50%RH
24 GHz	2.92	±0.06	0.0038	±0.0004	Split-post	23C/50%RH
40 GHz	2.91	±0.07	0.0040	±0.0005	Fabry-Perot	23C/50%RH
60 GHz	2.89	±0.08	0.0043	±0.0005	Fabry-Perot	23C/50%RH
77 GHz	2.88	±0.08	0.0044	±0.0005	Fabry-Perot	23C/50%RH
110 GHz	2.85	±0.10	0.0049	±0.0006	Free-space	23C/50%RH
140 GHz	2.84	±0.12	0.0053	±0.0007	Free-space est	23C/50%RH

Table 3-1: Dk/Df across frequency. Panel 0.8 mm, 1 oz RTF Cu, conditioned E-24/23/50.

3.1 Dk vs. Temperature at 10 GHz

Temp (C)	Dk	Df	Notes
-55	3.02	0.0031	Low-temp stabilised
-25	3.00	0.0032	
23	2.98	0.0034	Reference condition
50	2.96	0.0036	
85	2.94	0.0038	
125	2.91	0.0042	Near Tg onset
150	2.87	0.0049	Above Tg softening

Table 3-2: Dk/Df temperature dependence at 10 GHz (cavity resonator).

4. Thermal Conductivity

Direction/Parameter	Value	Unit	Uncert.	Temp	Method
In-plane (X)	0.45	W/m·K	±0.04	25 C	ASTM E1461 LFA
In-plane (Y)	0.44	W/m·K	±0.04	25 C	ASTM E1461 LFA
Through-plane (Z)	0.38	W/m·K	±0.04	25 C	ASTM E1461 LFA
Through-plane (Z)	0.34	W/m·K	±0.05	85 C	ASTM E1461 LFA
Thermal diffusivity (Z)	0.22	mm²/s	±0.02	25 C	ASTM E1461
Specific heat Cp	1.05	J/g·K	±0.05	25 C	DSC ASTM E1356
Therm. resistance 0.1mm	0.26	cm²K/W	±0.03	25 C	Calculated

Table 4-1: Thermal conductivity and related parameters (laser flash analysis).

5. Mechanical Properties (Extended Dataset)

Property	Symbol	Value	Unit	Uncert.	Standard
Flexural modulus (X)	E_fx	18.5	GPa	±1.2	IPC-TM-650 2.4.4
Flexural modulus (Y)	E_fy	18.2	GPa	±1.2	IPC-TM-650 2.4.4
Flexural strength (X)	sigma_f	320	MPa	±25	IPC-TM-650 2.4.4
Tensile modulus (X)	E_t	16.8	GPa	±1.0	ASTM D638
Tensile strength	sigma_u	285	MPa	±20	ASTM D638
Compressive strength	sigma_c	410	MPa	±30	ASTM D695
In-plane shear modulus	G_xy	4.2	GPa	±0.4	ASTM E143
Poisson ratio (X-Y)	nu_xy	0.18	—	±0.02	ASTM E132
Fracture toughness KIC	K_IC	0.85	MPa m^0.5	±0.08	ASTM E399
Fracture toughness KIIC	K_IIC	0.62	MPa m^0.5	±0.07	ASTM E1820
Hardness (Rockwell M)	H_R	88	HRM	±3	ASTM D785
Impact strength (Izod)	a_n	28	kJ/m²	±3	ASTM D256
Peel strength (1oz Cu)	P_s	1.45	N/mm	±0.05	IPC-TM-650 2.4.8
HAD (hole-wall accuracy)	HAD	<5	µm	—	IPC-TM-650 2.4.42

Table 5-1: Full mechanical dataset at 23 C unless noted.

6. Hollow Silica Microsphere — Particle Size Distribution

Laser diffraction (Malvern Mastersizer 3000) in isopropanol. Average of 5 production lots.

Parameter	Symbol	Value	Unit	Notes
D10 (volume)	D10	8.2	µm	10% of particles below
D50 (median)	D50	17.8	µm	Median particle diameter
D90 (volume)	D90	28.4	µm	90% of particles below
D97	D97	34.1	µm	Upper distribution cut-off
Span (D90-D10)/D50	Delta	1.13	—	Narrow = good uniformity
Shell thickness	t_s	1.2-1.8	µm	SEM cross-section
Shell/diameter ratio	t/D	~0.08	—	Target <=0.10
True shell density	rho_s	2.20	g/cm3	Solid fused silica ref.
Internal void fraction	f_v	0.85	—	Volume of air per sphere
Surface area (BET)	S_BET	3.2	m2/g	After APS functionalisation
Moisture (as-received)	w	<0.3	wt%	Karl Fischer titration

Table 6-1: HSM-G3 hollow silica microsphere PSD and morphology data.

7. Laser Absorption Coefficients & Ablation Parameters

Wavelength	Medium	Abs. Coeff.	Threshold	Rate	Via Quality	Notes
266 nm	MVC composite	4800 cm-1	0.40 J/cm2	18 um/shot	Excellent	UV; clean walls
355 nm	MVC composite	2200 cm-1	0.80 J/cm2	12 um/shot	Very good	UV DPSS Nd:YAG
532 nm	MVC composite	820 cm-1	2.10 J/cm2	10 um/shot	Good	Some melt zone
1064 nm	MVC composite	310 cm-1	5.80 J/cm2	6 um/shot	Fair	HAZ ~8 um
10.6 um CO2	MVC composite	9500 cm-1	0.15 J/cm2	8 um/shot	Good	Cu pre-drilled
355 nm	SiO2 shell only	600 cm-1	1.50 J/cm2	—	—	Inclusion only
355 nm	E-glass fabric	1400 cm-1	1.10 J/cm2	—	—	Reinf. material

Table 7-1: Laser absorption coefficients (photothermal deflection spectroscopy). Ablation rates at 2x threshold.

Ablation threshold: $F_{th} = E_p / (\pi * r^2)$. Recommend $F = 2.0\text{-}2.5 \times F_{th}$ for clean walls.

8. Benchmark: MVC V5.0 vs. Commercial Laminates

Property	MVC V5.0	Rogers RO4003C	Isola I-Speed	PTFE	FR4 IT-180A
Dk @ 10 GHz	2.98	3.55	3.45	2.10	4.20
Df @ 10 GHz	0.0034	0.0027	0.0040	0.0009	0.019
CTE X ppm/C	14.2	11	13	120	16
Tg (C)	185	>280	170	—	175
Lambda W/m·K	0.42	0.64	0.35	0.25	0.30
Density g/cm3	1.68	1.79	1.86	2.20	1.85
KIC MPa m^0.5	0.85	0.7	0.9	0.5	1.1
Cost index	1.0x	3.5x	2.2x	5.8x	0.4x
Open-source	Yes	No	No	No	No

Table 8-1: Competitive benchmark. Cost index normalised to MVC. Reference data from published manufacturer datasheets.